

Panithan Lertsuntivit

(949) 302-3234 | Panithan@berkeley.edu | [linkedin.com/in/panithan-lertsuntivit/](https://www.linkedin.com/in/panithan-lertsuntivit/) | panithan-lertsuntivit.github.io

EDUCATION

University of California, Berkeley
B.S. Mechanical Engineering

Expected December 2025
GPA: 3.73

EXPERIENCE

Aerospace SAE at Berkeley
Director of Avionics

September 2024 - Present
Berkeley, CA

- Fabricated a thrust test rig, and conducted propeller and motor tests in the Hesse AEROLAB wind tunnel
- Implemented Molex connectors and integration geometries for secure electrical and physical connections; facilitating a sub-2-minute fully functional wing panel assembly on the flight line for the 2025 SAE Aero Design Competition

UC Berkeley Mechanical Engineering
Teaching Assistant

August 2024 - Present
Berkeley, CA

- Facilitating weekly lab sections and providing hands-on experience with electronic components and circuits
- Improving student comprehension of course concepts through weekly discussions and office hours
- Regularly worked with the professor and other teaching assistants to ensure the course progressed on schedule

National Tsing Hua University
Yang Lab for Biomechanics | *Research Intern*

June 2025 - August 2025
Hsinchu, Taiwan

- Promptly adapted to the field of cephalopod research and experimentation by studying their biomechanics
- Iteratively engineered a fluid mixing adapter to produce a stable ink mixture comprised of ink and mucus, and validated its effectiveness in reproducing different ink flow geometries through rigorous testing
- Modeled and coded a controllable fluid release system to build a framework for iterative ink pattern testing

PROJECTS

Real-Time Plotter

- Developed a wireless ESP32-based plotter using Espressif's IoT Development Framework (ESP-IDF), capable of drawing geometric shapes at a frequency of 65 Hz directed from a LabVIEW graphical user interface
- Optimized system efficiency through multitasking between real-time stepper motor actuation and wireless communication by utilizing state machines, timer callbacks and task prioritization

Mobile Desk

- Designed a portable desk in SolidWorks and created detailed drawings with appropriate tolerances and datums
- Conformed to geometric dimensioning and tolerancing (GD&T) ensuring fits between parts for integration
- Manufactured aluminum components using manual lathes and mills in the UC Berkeley Etcheverry Machine Shop

Wall Climbing Vehicle

- Designed a remote control vehicle capable of maneuvering on a vertical surface using SolidWorks and Onshape
- Researched and prototyped various wooden living hinges, achieving flexible laser-cut wood geometries and rapid production of a wooden vehicle chassis using an FLS Muse Core Laser Cutter

Dark Matter Star Model

- Computed Dark Matter clumping in various pressure and energy densities using the Tolman-Oppenheimer-Volkoff equations to reconstruct a Dark Matter self-interaction model, under the guidance of Professor William Spinella
- Presented Dark Star mass-radius curves and findings at the 2023 UCI Building Bridges Research Conference

TECHNICAL SKILLS & INTERESTS

Relevant Coursework: Embedded Microprocessor Based Systems, Network Analysis, Dynamics, Thermodynamics, Manufacturing and Design Communication, Introduction to Solid Mechanics

Programs: SolidWorks, AutoCAD, Onshape, Microsoft Office Suite, ESP-IDF, LabVIEW, Civil3D

Languages: Python, MATLAB, C